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Safety-Shielded Hierarchical Reinforcement Learning for Formula 1: A Move-Level Racing Driver and a Lap-Level PPO Race Strategist

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Code, configurations and logged results: https://github.com/ADI-KRISH/Hierarchical-Multi-Agent-RL-for-Formula-1

ABSTRACT Formula 1 racing couples two sequential decision problems on different clocks: a driver who must overtake and defend at the grip limit without contact, and a strategist who chooses pit laps, tyres and push levels under safety cars, rain and incidents. Rule-based plans cannot react to such events, and end-to-end RL drivers learn slowly and lack hard safety. We build a seeded Gymnasium simulator on 25 real circuits and train a two-level PPO hierarchy. Its novelty lies in (i) a driver that selects temporally extended race-craft moves executed by a limit-pace path planner and filtered by a bilateral predictive safety shield, (ii) a curriculum seeded with real 2021-2025 overtakes used as start states rather than imitation targets, and (iii) a strategist trained on a profile measured from the learned driver. Against rule-based baselines on five unseen circuits, the driver records zero contacts in 1,174 real situations and holds 273 of 460 defences versus 264, while pure soft actor-critic plateaued 17-20% off the limit lap. One strategist covering eight race cases wins at least as often as a rule-based plan in seven of them on unseen circuits, including 89.5% versus 66.5% in the rain and 38.5% versus 13% after a spin.

INDEX TERMS reinforcement learning, hierarchical control, PPO, safe RL, shielding, autonomous racing, race strategy, curriculum learning, Formula 1.

I. INTRODUCTION

A. BACKGROUND AND CONTEXT

Reinforcement learning (RL) studies how an agent should act in an environment to maximise long-run reward [1]. At each step t the agent observes a state s_t, draws an action a_t from its policy $\pi_\theta(a \mid s)$, and the environment returns a scalar reward r_t and a new state s_{t+1}. The problem is usually written as a Markov decision process $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, and the goal is a policy that maximises the discounted return $G_t = \sum_{k\geq 0} \gamma^k r_{t+k}$. Because the agent learns from consequences rather than labelled answers, RL suits problems in which an early choice only pays off (or fails) many steps later. Deep networks made it practical for large state spaces: deep Q-networks learned Atari games from pixels [2], search plus policy networks mastered Go [3], and multi-agent RL reached grandmaster level in StarCraft II [4]. The same tools now appear in robot control [5], automated driving [6], drone racing [7] and planning problems where a model of the future is cheap to simulate but hard to solve analytically.

Motorsport is a natural but demanding test bed for these methods. A racing car is driven at the edge of tyre grip, where a small error ends the lap, and the reward for a good braking point is felt seconds later. Racing against others adds interaction: the car ahead shapes your air flow, your line and your options, and a collision is costly for both. Above the driver sits a slower layer of decisions - when to pit, which compound to fit, how hard to push - whose consequences unfold over a whole race and depend on what rivals do and on random events such as safety cars and showers. The key domain challenges are therefore (a) long horizons with delayed reward, (b) hard safety constraints, (c) other agents whose behaviour changes the problem, (d) two decision rates that differ by two orders of magnitude, and (e) scarce real data: public F1 telemetry has no steering signal and positions are sampled at only a few hertz.

B. PROBLEM STATEMENT

We address the following sequential decision problem. Given a circuit, a car model with a friction-ellipse grip limit and tyre wear, and a field of rival cars, learn (1) a **Driver Agent** policy that every 0.5 s chooses a race-craft move (follow the racing line, pull out of the slipstream, take a lane beside the car ahead, cover, take the inside, feint and similar) so that the car laps near the physical limit, completes clean overtakes, and defends its position with **zero contacts**; and (2) a **Strategy Agent** policy that once per lap chooses whether to pit, which compound to fit, and a push level, so that the car finishes a full Grand Prix as early as possible and ideally wins, under safety-car, virtual safety-car, rain, spin and puncture events. Both policies must generalise to circuits never seen in training.

C. MOTIVATION

Race engineering still leans on hand-written rules and Monte-Carlo race simulation [8], [27]. A fixed plan such as "start on mediums, change to hards at 45% race distance" is easy to explain but ignores the state of the race: it does not bring a stop forward under a safety car, it cannot react to a spin, and it never undercuts deliberately. Supervised learning can copy what teams did [10], but the labels encode those teams' cars, rivals and risk preferences, and there is no label for "the decision that would have been better". On the driving side, optimal control computes excellent single-car laps [11], [12] but becomes hard to solve once another car's intent is unknown; end-to-end RL can learn superhuman lap times in a high-fidelity game [13], [14], yet the published systems needed large compute budgets and still had to be taught etiquette through carefully weighted penalties.

RL is attractive here because the simulator gives unlimited, cheap experience, the objective (finish first, cleanly) is easy to state as reward, and the learned policy reacts to whatever state it finds itself in. The remaining gaps that motivate this work are: how to make RL learn clean wheel-to-wheel racing without an industrial compute budget, how to use real race data that lacks control signals, and how to make a strategist's decisions consistent with what its own driver can actually do.

D. OBJECTIVES

1. Build a fast, seeded Gymnasium simulator on real circuit geometry with a friction-ellipse car, tyre wear, slipstream, dirty air and aerodynamic drag.
2. Design a Driver Agent whose actions are race-craft moves rather than raw pedals, so that exploration happens at the level of tactics.
3. Create reward functions that pay for progress and for overtakes that stick, and that charge for contact, unsafe proximity and lost places.
4. Keep learning stable and safe with a predictive safety shield, behaviour-cloned warm starts and real-race start states.
5. Beat rule-based baselines: a scripted tactician and line-holding defender for the driver, and a fixed one-stop plan for the strategist.
6. Approach the best achievable result: the optimal-line limit lap for pace and a 100% win rate for the strategist where the car allows it.

E. CONTRIBUTIONS

1. **A move-level driver policy on top of a physics-aware path planner.** The PPO driver chooses among 17 temporally extended options (racing line, pull-out, four lane offsets with normal or late braking, cover, inside, feint, switchback and others); a planner then drives the chosen path at the grip limit. This turns a continuous control problem in which pure soft actor-critic plateaued 17-20% off the limit lap into a tactical choice made on top of a planner that, driving alone, laps within 1.1% of the limit.
2. **A bilateral predictive safety shield.** Every candidate command of every car is accepted only if both cars could still fall back on a rule-of-the-road manoeuvre (the car behind brakes, the car ahead keeps its lane) without touching; otherwise the least intrusive safe command replaces it. Contacts on 424 held-out real starts fall from 20 to 0, and the trained driver keeps zero contacts on all 1,174 held-out real starts.
3. **A real-incident curriculum without imitation.** Passes and defences from real 2021-2025 races (FastF1 and OpenF1 timing) are mapped onto the simulator's circuits and used as episode start states and as a held-out test set, which avoids behaviour-cloning targets that the data cannot support.
4. **Driver-aware strategy learning.** The Strategy Agent is trained against a statistical profile measured from the learned driver itself - its pace gap and logistic pass and defend curves - so the strategist values track position as much as the real driver's overtaking ability warrants.
5. **A composite race-time-plus-win objective with a stay-out prior.** A sparse win-only reward does not learn from a random start (a random policy pits on three laps out of four and never wins); biasing the initial policy towards staying out, and adding a dense race-time term, makes the win objective learnable and gives a policy that wins 200 of 200 held-out races when the car is quickest.

F. FUNCTIONALITY DIAGRAM

II. LITERATURE REVIEW / RELATED WORK

Optimal control for racing. The classic way to drive fast is to compute a trajectory and track it. Liniger et al. used model predictive contouring control to race 1:43 cars [11], and Heilmeier et al. computed minimum-curvature racing lines and controllers for a full-size autonomous race car [12]; the TUM racetrack database released with that work provides the circuit geometry we use. Game- theoretic extensions add an opponent [15]. These methods give near-optimal laps and clear guarantees, but they need an accurate model, they are expensive to solve with interacting agents, and tactics such

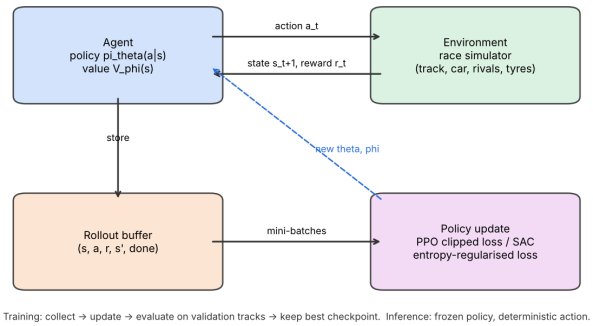


FIGURE 1. The RL interaction and update loop. During training the agent's actions are sampled from the stochastic policy, transitions are stored, PPO (or SAC for the pure-control baseline) updates the networks, and a periodic evaluation on validation circuits keeps the best checkpoint. At inference the frozen policy acts deterministically.

as a feint or a late lunge must be hand-specified. A survey of autonomous racing summarises these trade-offs [9].

End-to-end RL drivers. Early work learned lane following on a real car within a day [16], trained steering and throttle from images in a rally game [17], and ran open competitions in the TORCS simulator [18]. Fuchs et al. reached super-human lap times in Gran Turismo Sport with soft actor-critic [13], and Song et al. added overtaking through a three-stage curriculum [19]. GT Sophy [14] beat champion human drivers head-to-head; it used a distance-based passing reward, penalties for any collision, a mix of training scenarios and a population of opponents, trained on many consoles for days. These results show RL can find both speed and tactics, but they rely on large compute and on high-fidelity simulators, and their low-level policies have no explicit safety layer.

Hierarchical and multi-agent racing. Thakkar et al. split head-to-head racing into a high-level discrete game planner and a low-level controller learned with multi-agent RL, and won over 90% of races while obeying overtaking rules better than end-to-end MARL [20]. The general idea of temporally extended actions comes from the options framework [21] and later learned hierarchies such as option-critic [22], FeUdal networks [23] and HIRO [24]. In our design the options are hand-built race-craft manoeuvres and the hierarchy also spans the strategy layer.

Safe RL and shielding. García and Fernández surveyed ways to keep RL inside safe states [25]; Alshiekh et al. introduced shields that override unsafe actions of a learning agent [26]. Racing work usually relies on reward penalties instead [14]. Our shield is predictive and symmetric: it checks a short horizon for both cars and assumes each falls back on a rule-of-the-road manoeuvre.

Race strategy. Heilmeier et al. built a Monte-Carlo race simulation for circuit motorsport [8] and a "virtual strategy engineer" whose neural networks were trained on past decisions [10]. Thomas et al. trained an RL strategist (RSRL) for F1 and reported an average finish of P5.33 against P5.63 for the best baseline on the 2023 Bahrain Grand Prix [27]. Fieni et al. trained multi-agent strategists with self-play that manage energy, tyres and pit stops [28]. These works model the driver as a fixed lap-time function; none connects the strategist to the measured ability of a learned driver, and few test robustness to safety cars, rain and incidents at the same time.

Curriculum and data use. Curriculum learning orders tasks from easy to hard [29]; reward shaping changes the learning signal without changing the optimal policy when it is potential-based [30]. Behaviour cloning [31] provides warm starts but needs action labels, which public F1 data lacks.

Summary of prior approaches. Table~1 places the main lines of prior work side by side. Two patterns stand out. On the driving side, the strongest learned results [13], [14], [19] handle safety through penalties in the reward rather than through a component that can overrule an action, and they optimise lap time or overtaking with no strategist above them. On the strategy side, the learned strategists [10], [27], [28] take driving performance as a given lap-time model, so whether the driver can actually carry out the chosen plan lies outside their scope. Our system sits between these two lines and connects them.

Gap and positioning. Existing work either drives one car very well without an explicit safety guarantee, plans strategy with a fixed driver model, or needs large compute. Our approach is different in three ways: the driver acts at the level of manoeuvres on a planner that already laps near the limit, a shield enforces zero contact for every car including scripted opponents, and the strategist is trained on numbers measured from that same driver. This combination is necessary because, as our experiments show, a flat continuous-control agent did not reach competitive pace on our compute budget, and a strategist that assumes better overtaking than its driver has would choose plans the driver cannot execute.

III. SOLUTION APPROACH / METHODOLOGY

The pipeline has two learning levels that share one simulator family. At the lower level, `RaceEnv` integrates a car in the Frenet frame of a real circuit at 50 Hz; `TacticalEnv` wraps it so that the Driver Agent picks a manoeuvre every 0.5 s, a `PathDriver` turns the manoeuvre into a speed and lateral-offset plan, and the safety shield checks every car's command before the physics step. At the upper level, `StrategyEnv` simulates a full Grand Prix lap by lap for ten cars, using a lap-time table and pass/defend probabilities measured from the trained driver. Both agents are trained with PPO in Stable-Baselines3, evaluated on validation circuits during training, and scored once at the end on held-out circuits. The novelty sits in the move-level action space, the shield, the real-start curriculum and the measured driver profile that links the two levels.

A. BLOCK DIAGRAM

B. ENVIRONMENT DESIGN

TABLE 1. Positioning of related work

Work	Decision level	Method	Safety handled by	Opponents
Liniger et al. [11]	Driving	Model predictive contouring control	Constraints in the optimiser	Scale cars
Heilmeier et al. [12]	Driving	Minimum-curvature line and tracking control	Controller limits	None
Fuchs et al. [13]	Driving	Soft actor-critic, end to end	Reward penalties	None (time trial)
Song et al. [19]	Driving	Soft actor-critic with a curriculum	Reward penalties	Scripted cars
Wurman et al. [14]	Driving	Off-policy deep RL, scenario mix	Reward penalties	Human and AI drivers
Thakkar et al. [20]	Driving	Game planner over a multi-agent RL controller	Rules in the planner	Head-to-head rivals
Heilmeier et al. [8]	Strategy	Monte-Carlo race simulation	Not applicable	Simulated field
Heilmeier et al. [10]	Strategy	Neural networks trained on past decisions	Not applicable	Simulated field
Thomas et al. [27]	Strategy	Deep RL strategist	Not applicable	Simulated field
Fieni et al. [28]	Strategy	Multi-agent RL with self-play	Not applicable	Learned strategists
This work	Both	PPO over race-craft options; PPO strategist on a measured driver	Bilateral predictive shield	Scripted, RL-trained and frozen-self cars; rule-based field

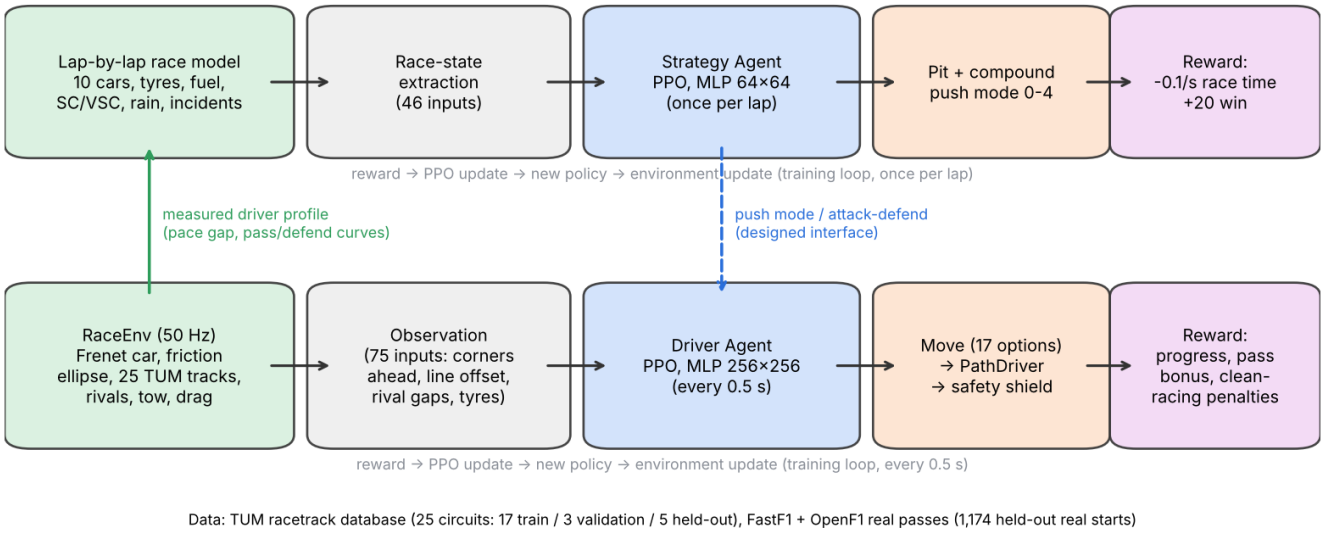


FIGURE 2. Environment → state extraction → agent → policy/value network → action → environment update → reward → training loop, shown for both levels. The green arrow is the measured driver profile that parameterises the strategy level; the dashed blue arrow is the push-mode interface from strategist to driver (designed, with the mode currently applied as a lap-time scaling in the strategy simulator).

1) Track, car and physics

Circuits. Twenty-five real circuits from the TUM racetrack database [12] give centre lines and track widths. They are split once: 17 for training, 3 for model selection (Montreal, Hockenheim, Catalunya) and 5 for the final test only (Suzuka, Spa, Sao Paulo, Silverstone, Yas Marina). A minimum-curvature racing line and a g-g (friction-ellipse) lap simulation give each track's limit lap, which serves as the pace reference.

Car. The car is a point mass with heading in the Frenet frame (arc length s, lateral offset n, heading error ψ). Longitudinal and lateral accelerations share one friction ellipse, $(a_x/a_{x,\max})^2 + (a_y/a_{y,\max})^2 \leq \mu^2$, with $a_{y,\max} = 5g$, braking $5g$, traction $1.5g$, top speed 97 m/s and an optional drag term chosen so that full throttle exactly holds top speed. Grip μ falls with tyre wear, which grows with tyre load and time.

Interaction effects. A car within a short distance behind

TABLE 2. Main simulator constants (R = reported F1 figure, D = design choice)

Quantity	Value	Source
Car mass	798 kg	R: FIA minimum weight
Peak lateral / braking / traction acceleration	5 g / 5 g / 1.5 g	R: reported F1 figures
Top speed	97 m/s (about 350 km/h)	R
Car length and width	5.5 m and 2.0 m	D
Physics step / control step / move decision	0.02 s / 0.1 s / 0.5 s	D
Slipstream range and drag reduction	80 m, 12%	D
Dirty-air range and grip loss	25 m, 4%	D
Tyre grip against wear w	$1-0.10w-0.15w^6$	D
Shield horizon and time step	2.5-3.0 s, 0.05 s	D
Shield margins along / across the track	1.0 m / 0.3 m	D
Pit-stop time loss: green / SC / VSC	22 s / 12 s / 14 s	D
Race distance	fewest laps covering 305 km	R: Sporting Regulations
Fuel at the start	110 kg	R: Technical Regulations

another gets a slipstream (12% less drag and a higher speed cap) but loses 4% grip in dirty air; all these values are design choices labelled as such in the configuration rather than claimed F1 figures.

Opponents. Scripted kinds include backmarker, slow, equal, defender, attacker and the "guard" defenders that make one legal move to the inside before a braking zone; RL-trained defenders and frozen copies of our own driver are also used.

Pace reference. Fig.~3 shows the optimal line on two held-out circuits. The model car reaches its 5 g limits in every corner and has no load transfer, so its limit lap is quicker than a real F1 lap (76.2 s at Suzuka). All pace results are therefore reported as a gap to this limit lap in the same physics, never as absolute lap times compared with real races. Table~2 lists the main constants and marks which are reported F1 figures and which are our own design choices.

2) State space

Driver Agent (75 inputs, all scaled to roughly $[-1, 1]$). Speed, lateral offset and heading; the corner speed limits in 12 look-ahead windows from 10 m to 250 m; the optimal line's offset and speed ahead (reference features); tyre wear and push level; gap, relative speed and lateral offset of the nearest car ahead and behind; the one-hot code of the move currently running; the corner after next and the metres left in the battle; and how the car ahead moved over the last second.

Strategy Agent (62 inputs). Race progress and laps left, place and gaps to the cars around, own compound, wear and fuel, whether the two-compound rule is met, the lap-time delta of each fresh compound against the current tyres, tyre life left against laps to go, the pit window per compound, where a stop would rejoin, rivals' compounds, tyre ages and last-lap pace, the track wetness and its forecast, safety-car and VSC flags, incident flags (spun, punctured, time lost, must stop) and the charge of the overtake-mode battery.

Table~3 breaks the final strategist's 62 inputs into groups; the earlier dry-weather configurations used the first 46. Every input is scaled to $[-1, 1]$, and several of them are computed answers to questions a race engineer would ask (where would a stop now rejoin, could a fresh tyre reach the flag) rather than raw state. In early runs without these derived inputs the agent did not discover the undercut, because the relation between a stop and the resulting position had to be learned from the raw gaps alone.

3) Action space

Driver. `Discrete(17)`: `line`, `outside`, `divebomb`, `dummy`, `switchback`, `cover`, `inside`, `pull_in`, `pull_out`, and eight lane moves (`left`/`right` of the car ahead at 3.0 m or 4.2 m centre-to-centre, each with normal or late braking). Once chosen, a lane move runs until the other car is cleared or 30 m behind (a committed option), which makes the problem a semi-MDP.

Strategist. `MultiDiscrete([6, 5, 2])`: pit choice (stay out, soft, medium, hard, intermediate, full wet), push level (five modes from 92% to 95% of grip, with tyre wear from 0.58× to 1.73× of normal) and overtake mode on or off. One lap in overtake mode uses half of the battery, which recharges by a tenth per lap, and adds 1.0 s to the car's pass advantage when attacking (0.5 s when defending). The first action picks the starting tyre for free. The earlier dry-weather configurations used only the first four pit choices and no overtake mode.

Stop rules. A few hard rules keep the strategist's choices realistic, in the same way the shield does for the driver. No new stop is allowed within five laps of the last one, and no stop by choice is allowed in the first quarter of the race unless the race is neutralised, the tyres are past 70% wear, the weather changes or the car is damaged. In the final configuration, voluntary stops in the last four laps and wet-weather tyres on a dry track are also blocked. These last two rules were added after training and are applied only at evaluation (Section 4.5).

4) Transition dynamics

The driver environment integrates all cars with a 0.02 s step; each agent decision repeats for five physics steps (0.1 s) and a move is reconsidered every 0.5 s. Before each step the shield simulates a 2.5-3.0 s horizon for each pair of cars and replaces any command that cannot be followed by a safe fallback. Episodes end on leaving the track, driving the wrong way, falling below 3 m/s, a 90 s time limit, or at the end of a section battle.

The strategy environment advances one lap per step. Each car's lap time is the measured lap for its grip level and fuel load, plus wear, push and dirty-air terms, plus pit loss when

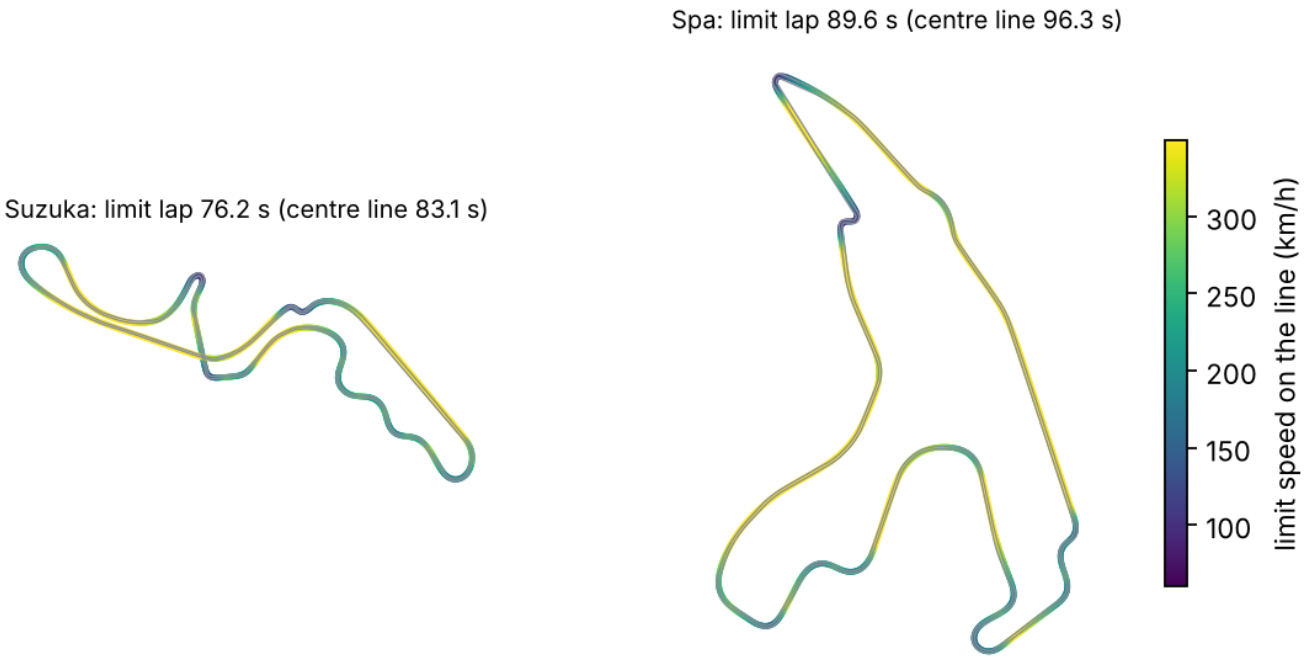


FIGURE 3. Minimum-curvature line on Suzuka and Spa, coloured by the limit speed the friction ellipse allows at each point; grey lines are the track edges. In the same physics, the optimal line is 6.9 s quicker than the centre line at Suzuka and 6.7 s quicker at Spa.

stopping (22 s; 12-14 s under SC/VSC). When a quicker car meets a slower one, a pass happens with a probability given by a logistic curve of the pace difference, fitted to the driver's measured battles; otherwise the quicker car follows 0.3 s behind. Rivals run seeded variants of the rule-based plan.

5) Reward shaping

Driver. Per decision,

$$\begin{aligned} r_t = \frac{\Delta s}{d_p} &- c_{\text{fail}}\,\mathbb{1}[\text{off/stall}] \\ &+ B\left(\mathbb{1}[\text{pass held}] - \mathbb{1}[\text{place lost}]\right) \\ &+ w_g\,\Delta g\,\mathbb{1}[\text{near}] - c_d\,n_{\text{dirty}} - c_i\,n_{\text{shield}} \\ &- c_w(1-p)\,\Delta w \pm r_{\text{section}}, \end{aligned}$$

with progress scale $d_p = 50$ m, failure penalty $c_{\text{fail}} = 50$, overtake bonus $B = 100$ paid only once a new order has held for 600 m (so passing and being re-passed earns nothing), a GT Sophy-style gap term $w_g = 1$ per metre gained on a nearby car [14], a penalty $c_d = 1$ per physics step of "dirty" driving (less than 0.5 m of air side by side, chopping across a car's nose, tailgating), a penalty for each step the shield had to intervene, a tyre-wear cost scaled by push level, and ± 30 at the end of a section battle.

Strategist. $r = -0.1\,(T_{\text{lap}} - T_{\text{ref}})$ per lap, summing to minus 0.1 times total race time, plus a bonus for a legal win at the flag ($+30$ in the final configuration, $+20$ in the earlier ones) and -50 if the car never used two dry compounds in a dry race.

Design rationale. Each driver term answers a failure we observed. The progress term alone produced a fast but timid driver, so the pass bonus pays for the outcome we want. A bonus paid at the moment of the pass was exploited by passing and being re-passed, so it is paid only after the new order holds, and a lost place costs the same amount. The gap term gives a dense signal while the car closes on a rival, which a bonus that arrives tens of seconds later cannot. The dirty-driving and shield penalties discourage moves that are legal only because the shield rescues them. The strategist's race-time term is dense and equals the objective of finishing as early as possible; the win bonus breaks ties in favour of a plan that takes the lead, and the large rule penalty makes an illegal one-compound race never worth it. None of these terms is potential-based [30], so each can change the optimal policy; this is intended, because "finish early" and "finish first" are different goals, and the bonus states which one we prefer.

6) Tyre, fuel and weather models

Tyre wear $w \in [0, 1]$ grows with how hard the tyre is worked. In the driving simulator the wear rate is $\dot{w} = u^2/L$, where $u \in [0, 1]$ is the share of the friction ellipse in use and L is the tyre life, and grip scales every acceleration limit by

$$g(w) = 1 - 0.10\,w - 0.15\,w^6,$$

a steady fade followed by a late cliff (Fig.~4a). In the strategy simulator each dry compound has a new-tyre grip and a life: soft 1.00 and 1,900 s, medium 0.985 and 3,000 s, hard 0.97

TABLE 3. Observation of the final Strategy Agent (62 inputs)

Group	Inputs	Count
Race progress	Laps done and laps left, as shares of the race	2
Position	Place; gaps to the cars ahead and behind; gap to the leader	4
Own car	Compound (one-hot); tyre wear; tyre age; fuel; stops; last lap against the reference lap	8
Rules and costs	Two-compound rule met; each compound used so far; pit loss relative to a lap	5
Nearby rivals	Tyre wear, tyre age and stops (relative to ours) of the cars ahead and behind; compound of the car ahead	7
Undercut	Place and gaps to the neighbours if we stopped this lap	3
Tyre economics	Lap time a fresh soft, medium or hard would gain; our tyre pace against the neighbours'; tyre life left against laps to go; whether each fresh compound could reach the flag	9
Incidents	Spun; punctured; time lost; stop required	4
Neutralisation	Safety car; VSC; laps neutralised so far; current pit loss	4
Weather	Wetness now; raining; forecast 3 and 6 laps ahead; on intermediates or wets; lap gain of fresh intermediates or wets; share of rivals on wet tyres; wet tyres used; wet-tyre overheating	11
Rival pace	Last lap of the cars ahead and behind, of the leader and of the field median, against ours	4
Overtake mode	Battery charge	1

and 4,500 s of driving at medium push. A push level $p \in [0, 1]$ multiplies wear by $3^{\,p-0.5}$, so attacking for a lap spends tyre life that must be repaid later (Fig.~4b). These numbers are design choices that create a genuine trade-off between a quick, short-lived compound and a slow, durable one; they were not fitted to Pirelli data.

Fuel enters through a lap-time table. For each circuit the limit lap on the optimal line is simulated over a grid of grip levels and fuel loads, and the strategy simulator interpolates in this table, so a car speeds up as fuel burns off and slows down as its tyres fade. Each car's lap is then scaled by the measured pace gap of our driver on that circuit (Section 3.4).

Rain is a per-race wetness curve. A shower starts at a random point between 10% and 70% of the race and lasts 10-40% of it; the track wets quickly and dries slowly. Slick grip falls linearly with wetness ω as $1 - 0.6\,\omega$, intermediates peak at 0.88 at $\omega = 0.45$, and full wets reach 0.82 on a soaked track (Fig.~4c), so the crossover points lie near 30% and 70% wetness. Wet-weather tyres overheat and wear up to four times faster as the track dries. The strategist reads a noisy forecast of the wetness three and six laps ahead.

7) Opponents, events and scenarios

Table~4 lists the other cars and the race events of the final strategy configuration. Every random draw (the circuit, the real start, the opponent, when an event happens and how long it lasts) comes from the environment's seeded generator. A battle or race is therefore identified by its seed and can be replayed exactly, which is how all paired comparisons in Section 4 are made.

C. RL ALGORITHM

We use **Proximal Policy Optimization** (PPO) [32] for both agents. PPO suits the discrete and multi-discrete action spaces, it is stable with many parallel environments, and its on-policy updates tolerate the non-stationarity caused by opponents and by changing scenario mixes better than replay-based methods. For the pure continuous-control baseline we used **Soft Actor-Critic** (SAC) [33], which is the standard choice for continuous throttle and steering [13].

PPO maximises the clipped surrogate

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\big[\min\big(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\big],$$
$$\rho_t = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)},$$

with advantages from generalised advantage estimation [34] $\hat{A}_t = \sum_{l\geq 0}(\gamma\lambda)^l\delta_{t+l}$, $\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$. The full loss is

$$L = -L^{\text{CLIP}} + c_v\,\mathbb{E}_t\big[(V_\phi(s_t) - \hat{G}_t)^2\big] - c_e\,\mathbb{E}_t\big[\mathcal{H}(\pi_\theta(\cdot \mid s_t))\big],$$

minimised with Adam for several epochs over mini-batches of each rollout; $\epsilon = 0.2$. SAC instead maximises $\mathbb{E}[\sum_t r_t + \alpha\mathcal{H}(\pi(\cdot \mid s_t))]$ with twin Q-critics, a target network updated with rate τ, and an automatically tuned temperature α.

D. NOVEL COMPONENTS

Options on a planner (hierarchical action space). Rather than learning to steer, the driver chooses what to do and the `PathDriver` handles how: it plans corner speeds and braking points on the path it will actually drive, so every option is executed at the grip limit. Scripted with no sideways

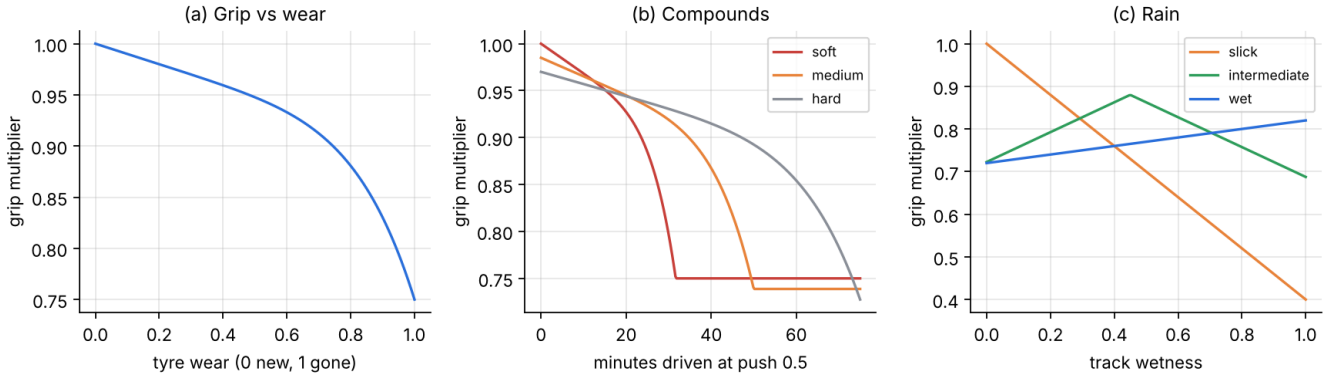


FIGURE 4. Tyre and weather models: (a) grip multiplier against tyre wear. (b) Grip of the three dry compounds against minutes of driving at medium push; the flat tail is a fully worn tyre. (c) Grip of slick, intermediate and full wet tyres against track wetness.

TABLE 4. Opponents and race events used in training and testing

Element	Level	Behaviour
Backmarker, slow and equal cars	Driver	Path driver on the racing line at a fixed share of the grip
Scripted defender and attacker	Driver	Rule-based race craft: cover the inside, attack from the slipstream
Guard defenders (slow, mid, equal)	Driver	One legal move to the inside before a braking zone, then hold the line
RL-trained defender	Driver	PPO policy trained to defend against our attacker
Frozen self	Driver	A copy of our own driver used as attacker or defender
Rule-based rival plans	Strategy	Medium-to-hard one-stop near 45% of the race, with seeded jitter of 5%
Spin or puncture	Strategy	In 35% of races; a spin costs 6-15 s, a puncture 15-30 s plus a forced stop
Safety car / VSC	Strategy	In 30% / 15% of races; SC 2-6 laps, VSC 1-3 laps; laps 40% / 35% slower; stops cost 12 s / 14 s instead of 22 s; a second period follows in 30% of these races
Red flag	Strategy	In 5% of races; every car may change tyres for free; attackers gain 0.5 s at the restart
Rain	Strategy	In 25% of races, as described in Section 3.2.6

move, this planner laps 24 usable circuits at +1.1% (worst +1.5%) off the limit lap.

Bilateral predictive shield. For each pair of cars the shield simulates the proposed commands over the horizon and accepts them only if both cars could still fall back on the rules of the road. Otherwise it searches candidates in order of intrusiveness - lift before braking, then brake and/or move away - and never picks one that leaves the road unless nothing else is clear. Four successive fixes were needed: treating the fallback itself as a candidate, preventing brake-tests, rejecting off-road swerves beside a car, and widening steering authority and horizon.

The procedure applied at every 0.1 s control step is summarised in Algorithm 2. Its key property is that the fallback it assumes is itself always one of the candidates, so a command judged safe now still leaves a safe command at the next step; this is what turns a local check into a guarantee over the whole battle, provided both cars are filtered.

Real-start curriculum. About 1,200 real passes and defences (FastF1 for 2021- 2022, OpenF1 for 2023 onwards) are converted to start states 400 m before the real pass location; 40% of training episodes begin from one. The held-out circuits' real starts form the final test.

Behaviour-cloned warm start. Before PPO, the driver is trained by supervised learning to copy the scripted tactician's choices (60k decisions, 95.9% agreement in one run; 40k defence decisions at 91.1% in the defend run), so exploration starts from competent play.

Driver profile and stay-out prior. The trained driver is measured on all 25 circuits (225 solo laps and 1,512 one-lap battles) to fit its pace gap and logistic pass and defend curves; the strategy simulator uses these numbers. When the win reward is sparse, the pit head's "stay out" logit is raised by 5 at initialisation.

The pass curves have the logistic form

$$P(\text{pass} \mid \Delta) = \frac{1}{1 + \exp\big(-(\Delta - \delta)/\sigma\big)},$$

where Δ is the attacker's pace advantage in seconds per lap. For our driver attacking, the fit over 648 battles gives $\delta = 4.9$ s and $\sigma = 1.71$ s; for rivals attacking our driver, 864 battles give $\delta = 1.0$ s and $\sigma = 1.15$ s (Fig.~6). The asymmetry is

```
Algorithm 2: Bilateral predictive safety shield for one pair of cars
Input: states x_A (car behind) and x_B (car ahead); proposed commands u_A, u_B
1:  H <- look-ahead of 2.5 s, extended up to 3.0 s when the cars close quickly
2:  function SAFE(x_A, u_A, x_B, u_B):
3:     integrate both cars over H: hold the commands for one decision,
        then apply the fallback (A brakes hard, B keeps its lane and pace)
4:     return the cars never come within one car length + 1.0 m along
        and one car width + 0.3 m across the track
5:  for the car behind first, then the car ahead (against A's final command):
6:     if SAFE(proposed command): keep it              # clean racing is untouched
7:     else try candidates in order of intrusiveness:
          car behind:          brake; brake and steer away; the fallback
          car beside or ahead: steer away; steer away and brake;
                               if it was braking, brake less (no brake-test)
8:        use the first candidate that is SAFE and stays on the road
9:     if none is SAFE: use the on-road candidate with the most room;
        leave the road only if that gives strictly more room
```

FIGURE 5. Bilateral predictive safety shield for one pair of cars.

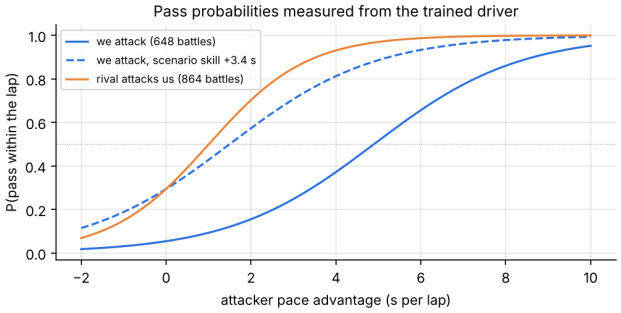


FIGURE 6. Probability of a pass within one lap against the attacker's pace advantage, fitted to battles of the trained driver on all 25 circuits. Solid blue: our driver attacking; orange: rivals attacking our driver; dashed: the attack curve with the 3.4 s skill offset used in the main strategy scenarios.

itself a finding: the learned driver defends well, but it needs almost five seconds a lap of extra pace for an even chance of passing within a lap. A strategist trained on this profile therefore learns to gain places in the pit lane rather than on track. The main strategy scenarios add an attack-skill offset of 3.4 s, which models a stronger overtaker; the measured curve itself is kept unchanged.

E. ALGORITHM DESCRIPTION

F. EVALUATION PROTOCOL

Four rules govern every number reported in Section 4.

Data separation. The 25 circuits are split once into 17 training, 3 validation and 5 held-out circuits. Training episodes use only training circuits, checkpoints are chosen on validation circuits, and held-out circuits are evaluated once after training ends, with no further tuning. The same split applies to the real-start library, so no real situation from a held-out circuit is seen during training.

Paired, seeded comparisons. A baseline and a learned policy always run on the same seeded episodes (same circuit, grid, rival plans and event draws), so a difference in outcome comes from the decisions alone. Held-out strategy scorecards use 200 races per case (seeds 70,000 to 70,199, circuits in rotation, 40 per circuit); driver scorecards use every real start or battle in the held-out set.

Strict success rules. A pass counts only if our car is ahead at the end, the new order has held for 600 m (1 km in the strict re-scoring), and there was no contact, no wheel off the track and no dirty-driving step. A defence counts only if the car is still ahead and drove cleanly. A race win counts only if the car used two dry compounds.

Statistics. For the three-seed study we report the mean, the sample standard deviation $\hat{\sigma}$ and a 95% confidence interval from the t-distribution with two degrees of freedom, $\pm 4.30\,\hat{\sigma}/\sqrt{3}$. For single runs we give the count and the number of trials, so a binomial interval can be computed.

IV. EXPERIMENTAL RESULTS

A. EXPERIMENTAL SETUP

Data. No offline RL dataset is used. Circuit geometry comes from the TUM racetrack database (25 circuits, pinned commit). Real overtaking situations come from FastF1 timing for 2021-2022 and from the OpenF1 API for 2023-2025; 2018-2020 were skipped because they now require a paid login. The held-out library contains 1,174 real starts on the five test circuits.

Environment. As in Section 3.2. Driver: 50 Hz physics, 10 Hz control, a move decision every 0.5 s, 90 s battles or 600-800 m sections spanning two corners. Strategy: full Grand Prix length (fewest laps covering 305 km), ten cars, our car starting tenth.

Hardware. All driver training ran on a laptop CPU (AMD Ryzen 7 7840HS, 8 cores / 16 threads, no GPU). Strategy runs ran on that laptop or on a 4-vCPU cloud container, also without a GPU.

Software. Python 3.12, Gymnasium [35], Stable-Baselines3 2.9 [36], PyTorch 2.14, NumPy, FastF1 3.8 and the OpenF1 REST API, with `uv` for reproducible environments. Every run writes its resolved configuration, seed and git commit next to its checkpoints.

Software practice. The physics, tyre, race and shield models are pure functions (state in, state out) with unit tests;

```
Algorithm 1: Hierarchical training of the Driver and Strategy Agents
Input: circuits split C_train / C_val / C_test; real-start library D; seeds
1:  // ---------- Driver level ----------
2:  init policy pi_theta, value V_phi (MLP 256x256); set seed
3:  pi_theta <- behaviour-clone(scripted tactician on C_train)        # warm start
4:  repeat until N_driver decisions:
5:     for each of K parallel envs:
6:        reset: pick circuit c ~ C_train; with prob 0.4 start from d ~ D(c),
                 else random section battle vs opponent from the mix
7:        observe s_t (75 features)
8:        a_t ~ pi_theta(.|s_t)                                      # move choice
9:        for each 0.1 s sub-step of the 0.5 s decision:
10:          u_i <- PathDriver(move_i) for every car i
11:          u_i <- Shield(u_i, others)                              # zero-contact filter
12:          integrate physics (dt = 0.02 s)
13:       r_t <- progress + held-pass bonus + gap term - dirty - shield - wear
14:       store (s_t, a_t, r_t, s_{t+1}, done)
15:    compute GAE advantages; for E epochs over mini-batches: Adam step on PPO loss
16:    every M decisions: evaluate on C_val; keep checkpoint if best and contact-free
17:  measure driver on all circuits -> profile P (pace gap, pass/defend logistic fits)
18: // ---------- Strategy level ----------
19:  init pi_psi (MLP 64x64); add stay-out prior to pit logits
20:  repeat until N_strategy laps:
21:     reset race: circuit ~ C_train, grid slot, rivals' plans, random SC/VSC/
                 spin/puncture/rain draws (seeded)
22:     each lap: observe race state (62 features); (pit, push, overtake) ~ pi_psi
         a stop that breaks the stop rules becomes stay out
23:        simulate one lap for 10 cars using P; r <- -0.1 * lap time delta
24:     at the flag: r += 30 * win - 50 * illegal
25:     PPO update as in lines 15-16, selecting checkpoints by validation wins
26: return pi_theta*, pi_psi*
Stopping: fixed decision/lap budget; best checkpoint chosen on C_val only;
          C_test is touched once, at the end.
```

FIGURE 7. Hierarchical training of the Driver and Strategy Agents.

the repository has 342 tests, plus type checking and linting, all run before every merge. Every environment, training run and evaluation takes an explicit seed, and no part of the simulator draws unseeded random numbers. Hyperparameters and reward weights live in one YAML file per experiment, and each run stores its resolved configuration, seed and git commit with its checkpoints, so any number in this report can be traced to the code and settings that produced it.

Development stages. The system was built in stages, each kept only once it ran and passed its tests (Table~5). The order matters for interpreting the results: the driver's action space was redesigned after the end-to-end SAC stage failed to reach competitive pace, and the strategist was started only after the driver could overtake and defend cleanly.

Hyperparameters.

Effect of each setting (observed in our runs).

- Discount factor. In an early lap-time driver, $\gamma = 0.98$ (a 5 s horizon at 10 Hz) learned to stop before a chicane because it could not see the value of the rest of the lap; $\gamma = 0.995$ fixed this. For the move-level driver, whose decisions last 0.5 s or a whole manoeuvre, 0.97-0.99 was enough.
- Control rate. At 50 Hz the early driver never discovered braking; 10 Hz worked; a 25 Hz SAC variant completed no lap after 250k decisions.
- Reward scale and learning rate. Multiplying all driver rewards by 10 sped up early SAC learning (+18% off the limit at 300k vs +26%), but linear learning-rate decay did not remove the final plateau. A smaller PPO learning rate (1e-4) in the defend run kept the attacking skill from being overwritten.
- Entropy and exploration. After behaviour cloning, entropy collapsed and the driver stopped attempting passes; raising the entropy coefficient from 0.01 to 0.03 together with a larger pass bonus restored them (Section 4.5). In self-play pilots the policy entropy still collapsed to about 0.3 nats.
- Batch size and epochs. Not swept systematically; values follow common PPO defaults scaled to the number of environments.
- Seeds. All earlier runs used seed 0. To measure seed sensitivity we retrained the main strategy configuration with three seeds (Section 4.2.5).

B. PERFORMANCE EVALUATION

1) Mean evaluation return and learning curves

Because the reward weights differ between runs, we report the task-level quantity each evaluation measured rather than raw return: lap-time gap for pure driving, success rate for battles, and win rate for strategy.

2) Success rate

Success is defined per task. A pass succeeds when our car is ahead at the end of the battle, the new order has held for 600

TABLE 5. Development stages and what each changed

Stage	What was built	Outcome
Simulator	Segment track, point-mass car, first Gymnasium environment	Physics as tested pure functions
Rule-based driver	Baseline driver and lap analytics	2.3-3.0% off the limit lap
First PPO driver	Throttle and steering on synthetic and six real circuits	Clean laps, 1-4% behind the baseline
Real circuits and SAC	25 TUM circuits, Frenet car, optimal line, SAC driver	Plateau 17-20% off the limit lap
Move-level driver	Path planner, race-craft moves, PPO picks moves	Planner at +1.1%; overtaking and defending possible
Real starts and shield	FastF1/OpenF1 starts, predictive shield	0 contacts on 1,174 held-out real situations
Passing physics	Slipstream, drag, shield in the tow	Scripted passes from 8 to 25 of 36
Section battles	Corner-by-corner battles, strict clean-racing rules	Learned driver level with the scripted tactician
Sparring partners	Guard defenders and an RL-trained defender	0 contacts in 480 guard battles
Defend run	Main driver also trained to defend	Held-out defences 273 vs 264, passes 49 vs 29
Strategy Agent	Lap-by-lap race model, PPO on the measured driver	72% wins vs 10%; rain 68% vs 0%
Every race case	Fast car from tenth, spins, punctures, SC/VSC	96.7% clean wins vs 91% (three seeds); safety-car weakness found
One agent for all cases	Stop rules, red flags, two SC periods, rain, rival pace, overtake mode	Wins or ties the plan in 7 of 8 cases; rain 89.5% vs 66.5%

TABLE 6. Training hyperparameters of the four learning set-ups

Setting	SAC driver (baseline)	PPO driver (defend run 2)	PPO driver (real starts)	PPO strategist
Learning rate	3e-4 $\rightarrow$ 3e-5 (linear)	1e-4	3e-4	3e-4
Discount γ	0.995	0.97	0.99	0.995
GAE λ / SAC τ	$\tau = 0.005$	0.95	0.95	0.95
Batch size	256	256	512	256
Rollout / buffer	1M replay	128 steps $\times$ 12 envs	256 $\times$ 8 envs	512 $\times$ 4 envs
Epochs per update	4 gradient steps / vec-step	8	6	10
Exploration	auto entropy temperature	entropy coef. 0.01 + BC warm start	entropy coef. 0.03 + BC	entropy coef. 0.01 + stay-out prior
Network	256×256 ReLU	256×256 ReLU	256×256 ReLU	64×64 tanh (SB3 default)
Training budget	2.0M decisions	0.5M decisions	1.5M decisions	1-4M laps (≈17k-67k races)

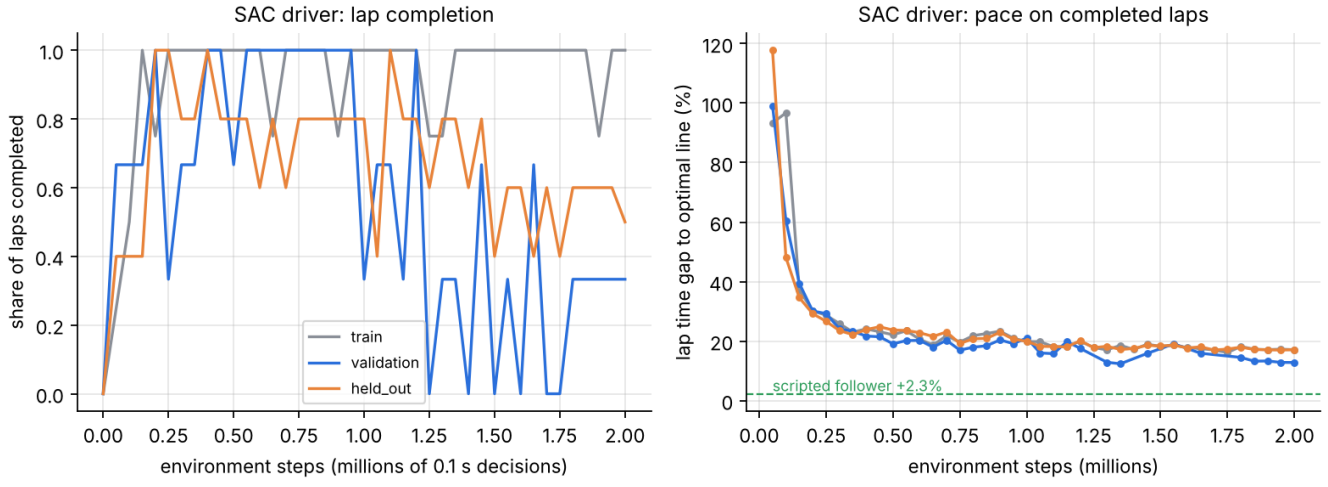


FIGURE 8. Pure-RL SAC driver (seed 0). Left: share of evaluation laps completed; right: mean gap to the optimal-line lap on completed laps. Pace improves quickly from +100% to about +25% within 0.3M decisions, then flattens at +13% to +20%; the scripted planner sits at +2.3%.

m (1 km in later re-scoring), and there was no contact, no off-track and no dirty step. A defence succeeds when the car is still ahead and clean. A race succeeds when the car wins legally.

*Later found to be inflated by place-swapping; see Section 4.3.



FIGURE 9. Strategy Agent validation win rate for five scenario configurations (30 races per point; 40 for the final one-agent configuration, which mixes every race case). Dotted lines are the rule-based plan on the same races.

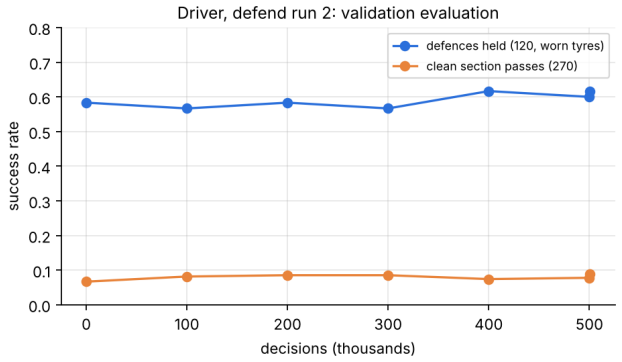


FIGURE 10. Driver defend run 2: validation defences held (120, worn tyres) and clean section passes (270) over 0.5M decisions. Both curves are nearly flat: the behaviour-cloned start already sits close to what these moves allow.

TABLE 7. Success rates on the held-out circuits

Task (held-out circuits)	Proposed	Best baseline
Defences held, worn tyres (460)	**273 (59.3%)**	264 (57.4%) line-holding
Clean section passes vs slow + guard cars (350)	**49 (14.0%)**	29 (8.3%) starting policy
Attack vs equal-pace defender, real starts (495)*	434 (87.7%)	168 (33.9%) scripted tactician
Race wins, clean, 1% faster car (200, final agent)	**100%**	99% rule-based plan
Race wins after a spin (200)	**38.5%**	13.0%
Race wins, safety car (200)	**95.5%**	93.5%
Race wins in the rain (200)	**89.5%**	66.5%

The strategy rows above are from the final Strategy Agent (`strategy_master`), a single agent trained once (seed 0, 4M laps) on a mix of every race case and scored on 200 held-out races per case with the stop rules of Section 3.2.3 switched on. Table~8 gives its full scorecard. It wins at least as often as the rule-based plan in seven of the eight cases

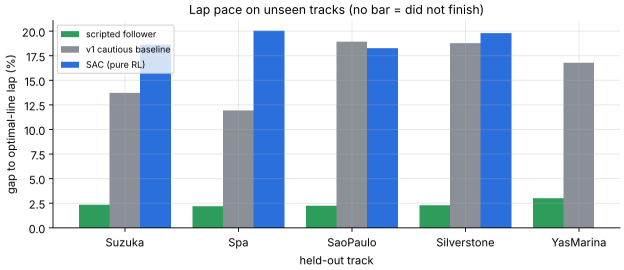


FIGURE 11. Gap to the optimal lap on each held-out circuit.

and finishes each case on average sooner; the exception is the VSC, where it wins 91% against 96%.

3) Cumulative regret

We define regret against the best available reference policy.

Driving (cost minimisation). Per lap, regret $= T_\pi - T_{\text{opt}}$, the time lost to the optimal-line limit lap in the same physics. Summed over the four held-out circuits that the SAC agent finished, cumulative regret is 57.9 s for SAC, 46.5 s for the v1 cautious baseline, and 6.8 s for the scripted path follower; Yas Marina is excluded because SAC did not finish it (scripted: 2.5 s there).

Strategy (reward maximisation). Per race, place regret is place $-$ 1 (distance from the best possible result). In the fastest-car, win-only scenario the agent's cumulative place regret over 200 held-out races is 0 (200/200 wins) against 22 for the rule-based plan (mean place 1.11). For race time we define regret per race as the time lost to the faster of the two policies on that seeded race. Over 200 clean held-out races the three seeds accumulate 44 s, 173 s and 26 s of race-time regret, against 1,125 s for the rule-based plan (measured against seed 2): the agent is the faster policy in almost every race. For the final agent we logged only per-case means: over its 1,600 held-out races (200 in each of the eight cases) it finishes 12,434 s sooner than the rule-based plan in total, 7.8 s per race on average, and its mean race time is lower in every case (Table~8).

4) Convergence speed and sample efficiency

We measure the number of environment steps until the validation win rate first exceeds the rule-based plan's, and until it reaches 90% of its own best.

The final configuration needs more experience than the others because it mixes eight race cases, rain and two extra actions (wet tyres and overtake mode) into one policy: it first beats the plan on validation after 1.0M laps and reaches 90% of its best win rate after 2.0M. The driver is far less sample-efficient: SAC needed about 0.3M decisions (8 h of simulated driving) just to finish laps reliably, and never matched the scripted planner within 2M decisions. The move-level PPO driver starts from a behaviour-cloned policy, so its first evaluation is already competent; the real- start run reached its attack plateau within 0.2M decisions (Fig.~14).

TABLE 8. Final Strategy Agent on held-out races, by case (agent / rule-based; 200 races per case)

Case	Wins	Mean place	Stops per race	Race time vs plan
Clean	**100%** / 99.0%	1.00 / 1.01	1.37 / 1.00	-6.6 s
After a spin	**38.5%** / 13.0%	3.42 / 5.24	1.52 / 1.00	-4.1 s
After a puncture	**5.5%** / 0.5%	8.06 / 9.70	1.92 / 1.64	-10.7 s
Safety car	**95.5%** / 93.5%	1.16 / 1.08	1.53 / 1.00	-6.7 s
Two safety cars	**94.5%** / 93.0%	1.14 / 1.15	1.77 / 1.00	-7.3 s
VSC	91.0% / **96.0%**	1.45 / 1.04	1.41 / 1.00	-3.5 s
Red flag	**98.0%** / 97.5%	1.08 / 1.02	1.73 / 1.60	-8.2 s
Rain	**89.5%** / 66.5%	1.58 / 2.56	2.56 / 3.07	-15.0 s

TABLE 9. Convergence speed and sample efficiency of the Strategy Agent (validation)

Strategy configuration	Baseline win rate	Steps to beat baseline	Steps to 90% of best	Best win rate
Fast car, win-only reward	93%	100k laps (≈1.7k races)	100k	100%
All cases, 3% faster car	93%	150k	100k	100%
All cases, 1% faster car	57%	400k (≈6.7k races)	450k	77%
Incident every race (wins)	0%	500k	1.1M	7%
One agent, every case (final)	50%	1.0M (≈17k races)	2.0M	70%

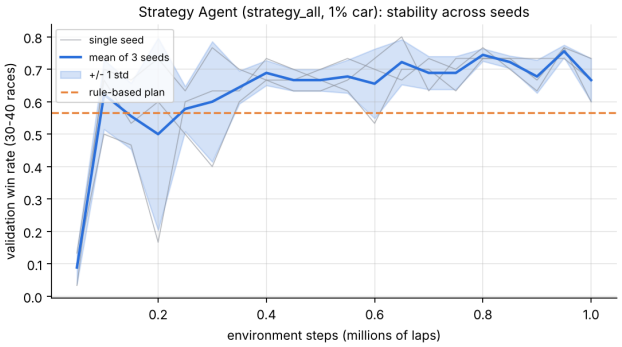


FIGURE 12. Validation win rate for three seeds: individual runs (grey), mean (blue) and one standard deviation (band). All three seeds pass the rule-based plan's 57% within 0.1-0.25M laps and settle at 60-80%.

5) Stability across seeds

Before the final configuration existed, we retrained the earlier all-cases configuration (`strategy_all.yaml`: 1% faster car, races with spins, punctures, SC and VSC) with three seeds (0, 1, 2) for 1M laps each, using identical code, and scored each seed's best checkpoint on the same 200 held-out races per case. These runs use the current configuration, which includes the minimum-stint rule (no new stop within five laps of the last).

Steps to first beat the rule-based plan on validation: 100k, 250k and 100k laps; best validation win rate 77%, 77% and 80%. The clean-race and after-spin advantages hold for every seed and their confidence intervals exclude the baseline. The safety-car result does not: every seed wins fewer SC races than the rule-based plan (78% vs 89.5%), although its mean race time is still 0.5-1.4 s shorter. The agent optimises race time, not position, and under the minimum-stint rule it makes extra stops under the safety car (1.4-1.6 stops against 1.0) that save time on average but sometimes cost the lead. The earlier single-seed agent trained without the minimum-stint rule won 94.5% of SC races, so this rule and the time-based reward interact; a position-aware term under neutralised running is the obvious fix. Seed 1 is the weakest in every case, which shows why a single seed should not be reported alone.

This study directly shaped the final configuration. To stop the agent from trading track position for small time gains, the final configuration forbids stops by choice in the first quarter of the race unless they are cheap or needed, models longer and repeated safety-car periods and red flags so that the agent sees many more of them, and raises the win bonus from 20 to 30. Under these settings the final agent wins 95.5% of safety-car races against 93.5% for the plan and 94.5% against 93% with two safety-car periods (Table~8). The final agent was trained with one seed only; repeating the three-seed study for it is the first item of future work.

6) Computational efficiency

A strategy training run used about 0.8 GB of resident memory in its main process; memory of the driver runs and energy use were not measured.

7) Problem-specific metrics

For racing we report: **gap to optimal lap (%)**, **contacts**, **off-tracks**, **dirty steps** (unsafe proximity), **shield interventions**, **clean passes held**, **defences held**, and **tyre wear per lap**. For strategy: **win rate**, **podium rate**, **mean finishing place**, **race time**, **legality** (two-compound rule), **stops per race**, **undercuts made and suffered**, and **on-track passes per race**. Generalisation is always reported on the five unseen circuits; robustness to opponents is reported against scripted, RL-trained and frozen-self opponents.

For the earlier all-cases configuration, whose three seeds give per-race data, Table~12 gives the strategy metrics behind the win rates of Section 4.2.5, and Table~13 splits the wins by circuit. Three patterns stand out. First, the agent's clean-race gain comes from the circuits where the rule-based plan sometimes loses (Suzuka, Silverstone, Yas Marina); where

TABLE 10. Stability across three seeds (held-out circuits, 200 races per case)

Held-out case (200 races)	Seed 0	Seed 1	Seed 2	Mean ± s.d.	95% CI	Rule-based
Clean race wins	97.0%	95.0%	98.0%	96.7 ± 1.5%	±3.8	91.0%
After a spin	33.5%	25.5%	30.5%	29.8 ± 4.0%	±10.0	4.5%
After a puncture	0%	0%	0%	0%	•	0%
Safety car	81.0%	78.0%	75.0%	78.0 ± 3.0%	±7.5	89.5%
VSC	93.5%	82.5%	86.0%	87.3 ± 5.6%	±14.0	89.0%
Mean race time vs rule-based, clean	-5.0 s	-3.4 s	-5.5 s	-4.6 ± 1.1 s		0

TABLE 11. Computational cost

Item	Value
SAC driver training (2.0M decisions)	3 h 29 min, laptop CPU, ≈165 decisions/s shared
PPO real-start driver (1.5M decisions)	3 h 31 min, ≈117 decisions/s with the shield on
PPO defend run 2 (0.5M decisions)	1 h 19 min, ≈104 decisions/s
PPO strategist (2M laps)	1M laps in about 15 min per run (three runs sharing 4 vCPUs, one thread each); ≈1.1k laps/s
Safety shield cost	≈35% lower environment throughput
Parameters	driver 175,122; strategist 14,986 (46 inputs) and 17,294 (final, 62 inputs)
Inference latency (CPU, batch 1)	driver 0.06 ms per decision; strategist 0.15 ms (46 inputs) and 0.21 ms (final) per decision (policy only)
Total logged training time	20 h CPU across all runs; no GPU used

the plan already wins every race (Sao Paulo) there is nothing to gain. Second, recovery after a spin varies widely by circuit, from 4% at Suzuka to 63% at Sao Paulo. A spin costs a similar number of seconds everywhere, but on Sao Paulo's short lap there are more laps left in which to recover it, which is the likely reason. Third, the safety-car weakness is concentrated at Suzuka and Spa, and it shows up in the stop count: the agent makes about 1.5 stops per safety-car race against exactly one for the plan.

C. RESULTS AND DISCUSSION

Result 1 - pure end-to-end RL plateaued. What happened: every SAC recipe (different look-ahead, reward scales, control rates, demonstrations, optimal-line features) plateaued at 17-20% slower than the limit lap; the best checkpoint finished four of five held-out circuits at +19.2%. Why: the agent found a cautious optimum - it failed only 2-5% of training episodes but averaged about 60 m/s against 70 m/s on the optimal line; one off-track costs 50 reward units, more than the time gained by braking later in many corners. Expected? Partly; published superhuman SAC drivers trained for far longer on richer signals [13]. Stability: completion on validation swung between 0/3 and 3/3 at the same average pace, so checkpoint selection was partly luck. Unseen tracks: held-out pace matched training pace, so the plateau was not overfitting. This result motivated the move-level design.

Result 2 - the shield removed contacts at small cost. With the scripted tactician on 424 held-out real starts, contacts fell from 20 to 0; off-tracks rose from 18 to 30 because the shield's swerves left the planner less margin, which later fixes reduced to zero on 284 validation starts with random moves. Every trained driver since then recorded zero contacts on held-out circuits except two, both caused by a scripted attacker at the same Silverstone corner. Component: the symmetric fallback check; reward penalties alone had left 1-4 contacts per evaluation.

Result 3 - reward exploitation in attacking, and its fix. With an overtake bonus of 100 the real-start driver beat the scripted tactician by a wide margin (434/495 vs 168/495 attacks). Watching replays revealed that the agent swapped places back and forth: it was paid for a pass that did not last. When passes had to hold 1 km over 150 s battles, the same model succeeded in only 7 of 49 attacks. We then (a) paid the bonus only once a new order held for 600 m and charged equally for re-passes, (b) added dirty-driving penalties, and (c) scored battles strictly. Under these rules the learned driver roughly matches but does not beat the scripted tactician on attacks (56 vs 65 of 525 held-out battles), while the defend-trained driver improves both defence (273 vs 264) and attack (49 vs 29 of 350). Where it failed: a brute-force oracle over the same moves found only 38 of 180 validation battles winnable within one section, so the ceiling, not the learner, limits overtaking at small pace gaps.

Result 4 - the strategist learned undercuts and event responses. Trained against rivals that all run the rule-based plan, the agent discovered a race-long undercut: a medium-to-hard stop at 36-42% of the race, earlier than every rival, then inheriting the lead as they stop later on older tyres. With our measured driver (which needs a large pace edge to pass) it won 72% of held-out races against 10% for the rule-based plan, and finished on the podium in 95% against 33%. In the rain it pitted about a lap earlier than rivals by reading the forecast (68% wins against 0%). After a spin it pits next lap for fresh tyres and attacks (29% wins vs 4.5%). Which component: the race-state features (tyre delta, pit window, rejoin position) and the race-time reward; the win-only reward did not learn without the stay-out prior.

TABLE 12. Earlier all-cases configuration: race statistics on held-out circuits (agent mean of three seeds / rule-based; 200 races per case)

Case	Wins	Podiums	Mean place	Stops per race
Clean	96.7% / 91.0%	98.3% / 99.0%	1.16 / 1.14	1.01 / 1.00
After a spin	29.8% / 4.5%	45.8% / 21.0%	5.18 / 6.64	1.42 / 1.00
After a puncture	0% / 0%	1.2% / 0%	9.55 / 9.94	2.01 / 1.65
Safety car	78.0% / 89.5%	87.2% / 97.0%	1.88 / 1.19	1.53 / 1.00
VSC	87.3% / 89.0%	92.2% / 98.0%	1.60 / 1.19	1.06 / 1.00

TABLE 13. Earlier all-cases configuration: held-out win rate (%) by circuit (agent mean of three seeds / rule-based; 40 races per circuit and case)

Circuit	Clean	After a spin	Safety car	VSC
Suzuka	94.2 / 82.5	4.2 / 0.0	62.5 / 82.5	79.2 / 82.5
Spa	93.3 / 92.5	17.5 / 2.5	70.8 / 82.5	77.5 / 87.5
Sao Paulo	100.0 / 100.0	63.3 / 17.5	95.8 / 97.5	98.3 / 100.0
Silverstone	96.7 / 87.5	24.2 / 2.5	80.8 / 90.0	89.2 / 87.5
Yas Marina	99.2 / 92.5	40.0 / 0.0	80.0 / 95.0	92.5 / 87.5

Where it failed: a puncture costs 40-50 s, which a 0.3 s/lap edge cannot recover (0% for both policies); VSC wins were level (88.5% vs 89%).

Result 4b - one agent for every case. The final agent handles all eight cases with one policy (Table~8, Fig.~15). In the rain it gains most: 89.5% wins against 66.5%, with fewer stops (2.56 against 3.07); unlike the plan, it can read the wetness forecast and the share of rivals already on wet tyres (Table~3). On Suzuka alone, 100 extra rain races (seeds 80,000 to 80,099) give 87% wins against 63%. After a spin it wins 38.5% against 13%, stopping more often than the plan (1.52 against 1.00 stops per race) to recover on fresh tyres. Where it failed: after a puncture it still wins only 5.5% of races, because the puncture costs more time than a 1% faster car can recover; under the VSC it often takes the cheaper stop and then loses time passing cars on track, and wins 91% against 96%. The last checkpoint of training was better after a spin (48% wins) but worse in the other cases, so selecting the checkpoint on validation wins mattered (Table~16).

Result 5 - generalisation. Every number above is on circuits never used for training or model selection. Strategy performance on held-out circuits was lower than on validation in the first pilot (place 4.20 vs 2.75), and similar in later runs; driver results transferred with no visible gap.

Result 6 - one race in detail. Fig.~13 follows a single held-out race at Suzuka (the first race of the safety-car scorecard). Panel (a) runs it with the same seed for the rule-based plan and for the seed-0 agent of the earlier configuration. Both start tenth on mediums and both stop for hards on lap 25. The agent drives most of its first stint at a lower push level (0.25 against 0.5), so its tyres are less worn at the stop (wear 0.44 against 0.58 on lap 20). When the safety car comes out on lap 35 the agent is third. It makes a second, cheaper stop on lap 36 for new hards and rejoins tenth, because under the safety car the field runs nose to tail and even a cheap stop costs every place. The rule-based plan stays out, moves up as the remaining rivals stop, leads from lap 42 and wins. With fresh tyres the agent climbs back to fifth but runs out of laps, and its race time is 7.1 s longer. This race shows the safety-car weakness of Section 4.2.5 in concrete form: the stop is cheap in seconds, but the lost track position cannot be recovered with the driver's measured overtaking ability (Fig.~6).

Panel (b) runs the same seed under the final configuration, whose safety car falls later (laps 41-45). The final agent starts on softs, conserves them for the first laps, leads from lap 16 and then pushes to build a gap. It stops for mediums on lap 23 (two laps before the plan) and leads again by lap 27. When the safety car comes out it is far enough ahead that a cheap stop for new mediums costs only one place; it retakes the lead on the second lap after the restart and wins, finishing 3.3 s sooner than the plan. The difference from panel (a) is not a cheaper stop but the larger gap before it, which the agent built with a quicker first compound and an earlier stop.

D. COMPARISON WITH EXISTING / BENCHMARK METHODS

Driver.

†Different battle set (two-corner battles against slow, guard and RL defenders). ‡Lap pace comes from the planner; under the full tactical layer the solo lap was +3.1%.

Strategist (held-out races, 1% faster car, starting tenth).

Results from other papers use different simulators and fields, so they are listed for context, not as a direct numerical comparison.

E. ABLATION STUDY

F. ROBUSTNESS, SENSITIVITY AND GENERALIZATION

Robustness (environment changes). Opponent change: the defending driver was tested against four attacker types, including a frozen copy of itself; the overtaking driver against slow, guard and RL-trained defenders. Across all of these opponents the shielded driver recorded two contacts in 810 held-out battles, both caused by the same scripted attacker at one corner. Against the RL-trained defender the learned attacker passed 14 of 175 times against 15 for the scripted tactician, so a stronger opponent lowers the success rate of both drivers alike. Failures and events: the final strategist was evaluated separately on eight race cases: clean races, spins, punctures, one or two safety cars, VSCs, red flags and rain (Fig.~15). It wins at least as often as the rule-based plan

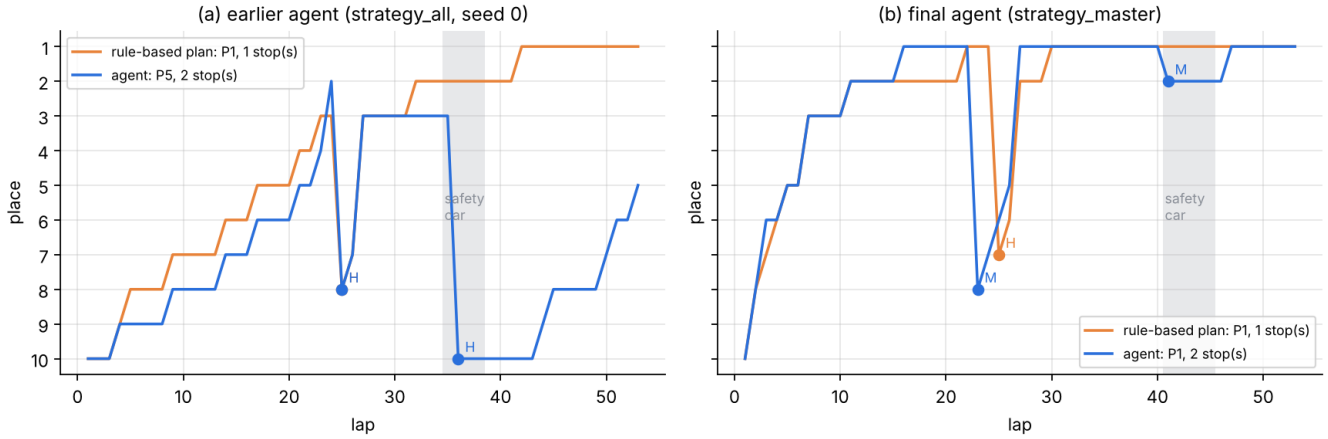


FIGURE 13. Place per lap in one seeded held-out race at Suzuka with a safety car (grey band): (a) the earlier agent, (b) the final agent, each against the rule-based plan on the same seed under its own configuration. Dots mark pit stops and the letter the new compound.

TABLE 14. Driver Agent compared with baseline methods (held-out circuits)

Method	Gap to optimal ↓	Clean passes (350) ↑	Defences held (460) ↑	Contacts ↓	Unseen circuits	Train time ↓
v1 cautious rule-based driver	+11.9% to +18.9%	•	•	•	5/5 finished	none
SAC end-to-end [33]	+18.3% to +20.1%	•	•	•	4/5 finished	3.5 h
Scripted path follower (no moves)	+1.1% (25 circuits)	0	264	0	5/5	none
Scripted tactician (rule-based moves)	+1.1%	65 / 525†	257	0	5/5	none
PPO move-level driver + shield (proposed)	+1.1% planner‡	**49** (start 29)	**273**	2	5/5	1.3 h (+ earlier stages)

TABLE 15. Strategy Agent compared with baseline methods (held-out races)

Method	Win rate ↑	Mean place ↓	Legal ↑	Unseen circuits	Decision time
Random policy	0%	10.0	95%	yes	•
Rule-based M→H one-stop at 45%	91% clean / 4.5% after spin (earlier set-up); 99% / 13% / 93.5% SC / 66.5% rain (final set-up)	1.11 (win-only scenario)	100%	yes	negligible
RSRL [27] (reported, different simulator)	•	P5.33 vs P5.63 baseline	•	Bahrain 2023	•
PPO strategist, earlier configuration (3 seeds)	96.7% clean / 29.8% after spin / 78% SC	1.00 (win-only scenario)	100%	yes	0.15 ms
PPO strategist, final (one agent, 8 cases)	**100% clean / 38.5% after spin / 95.5% SC / 89.5% rain**	**1.00** clean, 1.58 rain	100%	yes	0.21 ms

in every case except the VSC. In an earlier test with a 3% faster car, the strategist won 99.5-100% in every case except punctures (56% vs 44% for the rule-based plan). Forecast error: the rain agent reads a noisy 3- and 6-lap forecast and still pitted about one lap before rivals. Sensor error: not tested; observations are exact.

Sensitivity (parameter changes). With car pace edges of 0.5%, 1% and 3%, the agent's advantage over the rule-based plan is largest where the race is close (1%: 96% vs 91% clean, 29% vs 4.5% after a spin) and vanishes when the car is so fast that any sensible plan wins (3%). With a stronger overtaker

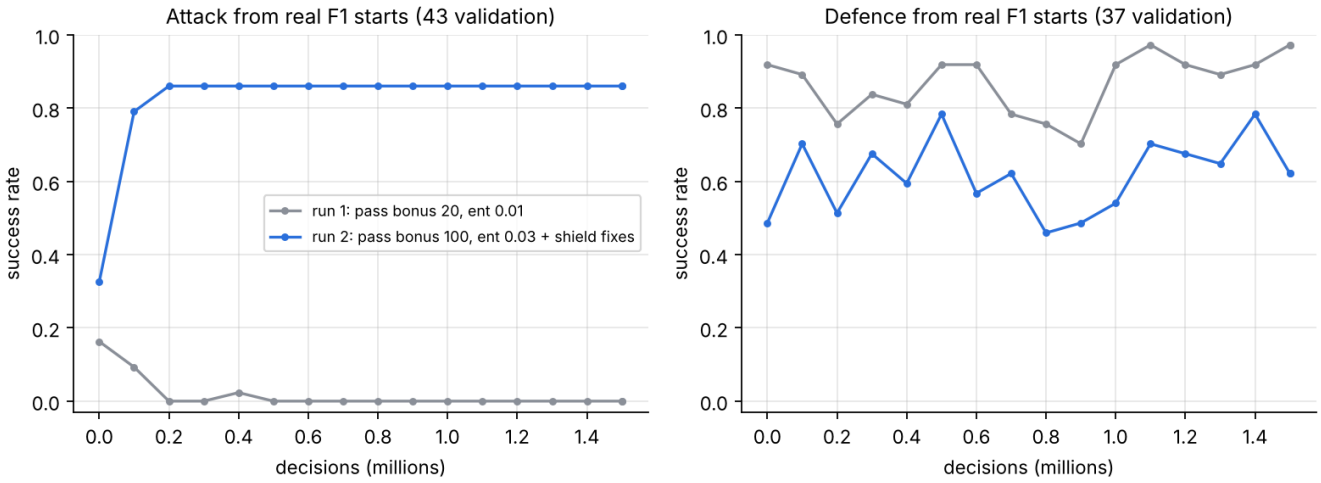


FIGURE 14. Real-start driver, run 1 (overtake bonus 20, entropy coefficient 0.01) vs run 2 (bonus 100, entropy 0.03, plus shield fixes). Run 1 stops attacking within 0.2M decisions; run 2 attacks on 37/43 starts but defends worse.

TABLE 16. Ablation study

Component removed or changed	Effect (validation unless noted)
Overtake bonus 100 → 20, entropy 0.03 → 0.01	attack success 37/43 → 0/43 after 0.2M decisions (passive policy)
Pass must hold 600 m / 1 km	without it, reported attacks 434/495 fell to 7/49 under strict scoring
Safety shield	contacts 0 → 20 on 424 held-out real starts (scripted driver)
Shield fixes 1-3 (fallback as candidate, no brake-test, no off-road beside car)	random-move contacts 2 → 1, off-tracks 26 → 3 on 284 starts
Shield fix 4 (steer 0.4 → 0.8, horizon 1.5 → 2.5 s)	18 contacts → 0 in 150 validation defences
Move-level actions → raw continuous control (SAC)	gap to limit +1.1% (planner) → +17-20%
Optimal-line features added to SAC	no gain (+19-21% vs +19.2%)
Air drag + two-corner battles	scripted passes 13 → 23 of 180; off-tracks 18 → 9
Dirty air removed (diagnostic)	scripted passes 23 → 36 of 180
Stay-out prior removed (win-only reward)	no learning: random policy pits 3 laps in 4, never wins
Dense race-time reward vs win-only (incident races)	win-only: P10, pitting every lap; dense: P6.9 vs 8.1 baseline
Driver profile: strategist trained on rain mix used in dry races	dry wins 25% vs 72% for dry-trained agent
Final agent: late-stop and dry-track wet-tyre rules switched off at evaluation (held out)	rain wins 92.0% vs 89.5% with the rules; other cases within 0.5 points; the rules remove unrealistic stops rather than add wins
Final agent: last checkpoint instead of the best on validation (held out)	after a spin 48% vs 38.5%, but two SCs 88% vs 94.5%, red flag 93.5% vs 98%, VSC 88.5% vs 91%
Earlier → final configuration (early-stop rule, longer and repeated SCs, win bonus 30; several changes at once)	safety-car wins 78% (three-seed mean) → 95.5%

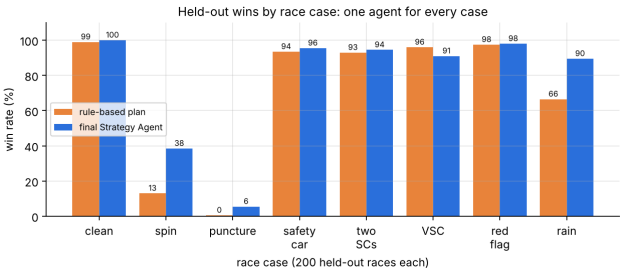


FIGURE 15. Held-out wins of the final Strategy Agent by race case, 200 races each, 1% faster car starting tenth, against the rule-based plan on the same seeded races.

(pass at a 1.5 s/lap edge) the agent switched from a clean-air plan to racing through traffic (11.4 vs 7.8 passes per race). Grip-to-lap-time sensitivity: 1% grip ≈ 0.3 s per lap.

Generalisation. Driver and strategist were selected on three validation circuits and tested once on five unseen circuits that differ in layout (Spa's long straights, Suzuka's esses, Sao Paulo's short lap). The held-out library of real starts adds unseen situations drawn from real races.

V. LIMITATIONS

- **Simulator fidelity.** The car is a point mass with a friction ellipse, no load transfer, no tyre temperature and simplified aerodynamics. Slipstream, dirty air, tyre compounds, pit loss, safety-car and weather effects are design values, not calibrated F1 figures; a one-race FastF1 tyre fit was not identifiable because tyre age and fuel burn are collinear.
- **Data limitations.** Public data has no steering or pedal signals and only ~4 Hz positions; 2018-2020 sessions now need a paid login; OpenF1 races have no pass

locations yet, so only their defences are used.

- **Overtaking ceiling.** At realistic pace gaps (0.4-1.2 s per lap) clean passes remain rare for both the learned and the scripted driver; the learned driver does not yet beat the scripted tactician on strict attacking.
- **Reward exploitation.** The place-swapping episode shows the reward can be gamed; the strict scoring fixes the known case but cannot rule out others.
- **Loose coupling of the hierarchy.** The strategist uses a statistical profile of the driver; the driver does not yet receive the push mode as an input, and the two levels are not fine-tuned jointly.
- **Rival behaviour.** Strategy rivals follow seeded rule-based plans and never react to an undercut, which flatters early stops; there is no cold-tyre out-lap.
- **Statistical scope.** Apart from the three-seed strategy study, results come from one seed per configuration, including the final Strategy Agent; the seed study itself found a safety-car weakness (78% vs 89.5% wins) that one seed alone had hidden. Several of the final agent's margins (safety car, two safety cars, red flag) are 0.5-2 points on 200 races and lie within binomial noise; validation sets of 30-40 races give wide confidence intervals.
- **Hand-written stop rules.** The final strategist relies on rules that block early, late and wet-on-dry stops. They keep the plans realistic but mean that part of its behaviour is specified rather than learned, and two of the rules were added after training.
- **Computational cost.** Driver training runs at 100-170 decisions per second on a laptop CPU, and the shield costs about a third of that, which limited the number of driver experiments and seeds.
- **Unresolved failures.** Punctures cannot be recovered with a small pace edge (5.5% wins); the final strategist loses more VSC races than the plan (91% vs 96%); the defend scenario on held-out circuits succeeded 2/5; lap pace under the full tactical layer is +3.1% instead of the 1% target.

Threats to validity. Internal validity: the baselines were written by the same team that designed the agents and the simulator. They were fixed before the held-out evaluation and never tuned on it, but a stronger hand-written plan, for example one that also brings its stop forward under a safety car, could narrow the strategy gaps. Construct validity: a win rate against a field of rule-based rivals measures how well the agent exploits those rivals, not how it would fare against teams that react to an undercut. External validity: tyre, pit and event constants are uncalibrated design values, so the learned plans (for example the timing of the undercut) should not be read as advice for real races. Conclusion validity: driver results rest on one seed, and the battle sets contain a few hundred episodes. The defence result (273 vs 264 of 460, 59.3% vs 57.4%) lies within binomial noise and should be read as "at least as good as" the line-holding baseline; the pass result (49 vs 29 of 350, 14.0% vs 8.3%) is about 2.4 standard errors apart and is more reliable.

VI. CONCLUSION AND FUTURE WORK

We studied racing as a two-timescale decision problem in which an F1 car must be driven fast and cleanly while a strategist manages tyres and pit stops through unpredictable events. The gap we targeted is that end-to-end RL drivers need large compute and lack hard safety, while RL strategists are trained on fixed lap-time models unrelated to any real driver. Our system combines a PPO driver that selects race-craft options on a near-limit path planner, a bilateral predictive shield that enforces zero contact, a curriculum seeded with real F1 passes, and a PPO strategist trained on the measured profile of that driver. The experiments establish four things: flat continuous control (SAC) plateaued at 17-20% off the limit while the planner-based hierarchy laps within 1.1-3.1%; the shield eliminated contacts across more than a thousand real held-out situations; the defending driver improved held-out defences (273 vs 264) and passes (49 vs 29); and the strategist learned undercuts and event responses that beat a rule-based plan on unseen circuits. Across three seeds of an earlier configuration the clean-race and after-spin gains were stable ($96.7 \pm 1.5\%$ and $29.8 \pm 4.0\%$ wins against 91% and 4.5%), while safety-car races exposed a weakness of the race-time reward (78% vs 89.5%). The final configuration, one agent for eight race cases with simple stop rules, removed that weakness (95.5% vs 93.5% under a safety car) and wins at least as often as the plan in seven of the eight cases, with the largest gains in the rain (89.5% vs 66.5%) and after a spin (38.5% vs 13%); it still loses more VSC races than the plan (91% vs 96%).

Three lessons from this project apply beyond racing. First, the choice of action space mattered more than the choice of algorithm: the same compute that left a continuous-control agent 17-20% off the limit produced a competitive driver once the agent chose manoeuvres and a planner executed them. Second, a score is not a behaviour: the place-swapping exploit was invisible in the learning curve and was found only by watching replays, and it was fixed by changing what counts as success, not by tuning the learner. Third, a single seed can hide a weakness: the safety-car problem appeared only when the same configuration was trained three times.

Future work follows from the limitations: calibrate tyre and pit constants from multi-race FastF1 stints; give the driver the push mode and attack/defend orders as inputs and fine-tune both levels jointly; replace scripted rivals with a league of learned opponents and reactive strategists, as in population-based self-play [4], [14]; add defensive late braking and holding the inside through a corner, which the oracle analysis identifies as the missing moves; add cold-tyre out-laps, fuel saving and lapped traffic; send the strategist's overtake-mode calls to the driver; and repeat the final strategist and every driver experiment over several seeds with confidence intervals [37], [38].

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